Optimization of Coil Design for Wireless Power Transfer System of 40W Incandescent Bulb

Ronith Kumar Devarakonda - 2020EE10486 Navya Gampala - 2020EE30595 Dr. Sumit Pramanick

***Abstract*—In wireless power transfer, designing efficient coupling coils is vital for minimal power loss in Incandescent Bulb applications. This involves achieving the target mutual inductance with minimal power loss by optimizing parameters like turns, size, and more.**

I. Introduction

To power a 40W Incandescent bulb wirelessly via Inductive Power Transfer, the setup includes transmitting and receiving coils. AC supply is sent through a Diode Bridge Rectifier on the transmitting side, then through an inverter, and finally to the primary coil. This method ensures frequency matching and safety. Resonant capacitors on both sides aid impedance compensation, maintain high frequency, and align current and voltage phases. The received power is again sent through rectifier before reaching the bulb. See Figure 1. We analyze

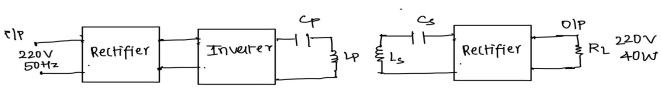

Fig. 1: Wireless Power Transfer System of 40W Bulb

the circuit to determine parameters for meeting a 40W, 220V DC output for a 220V AC, 50Hz input on the left side. Assumptions: Coupling coefficient K= 0.2, Frequency f= 85KHz. From analysing the circuit and applying KVL we get i_s = 0.1818A,R_L = 1210Ω, $M = 2.265 * 10^{-3} H, i_p = 0.186A$

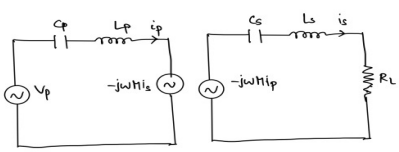

Fig. 2: Analysis of the circuit

II. Coil Parameters

II-ASymmetric Coils: Assume, the inner Radius of the coils R_{in}=0.075 m, Width of the coils W= 0.01 m and the spacing between each turn to be T=0.005 m. By observing we can see that inner diameter(D_{in}) and outer diameter(D_{out}) are in following relation

$$D_{\text{out}} = D_{\text{in}} + 2W + (T + W)(2N - 1) \tag{1}$$

Our initial assumption is that both coils are distant from each other and the flux density is constant throughout the coils. Using Biot-Savarts law,

$$M = \frac{\mu_0 \pi}{2} \sum_{i=1}^{Ns} \left(\sum_{j=1}^{Np} \frac{R_s^2}{(R_p^2 + d^2)^{\frac{3}{2}}} \right) \tag{2}$$

For symmetric case, we get N=N_p=N_s=23 to achieve a target mutual inductance M = $2.265*10^{-3}$H.

$$R = \rho * L \div A \tag{3}$$

$$L = 2\pi * n * (R_{\text{in}} + \frac{n-1}{2} * T + \frac{n+1}{2} * W) \tag{4}$$

Then the power losses in the coil is obtained from

$$Loss = {i_{\text{p}}}^2 R_{\text{p}} + {i_{\text{s}}}^2 R_{\text{s}} = 0.000522778W \tag{5}$$

III. Optimized Coil Parameters

For optimising the coil i.e; reduce the power losses

III-AAsymmetric Coils: Using $M = 2.265 \times 10^{-3}$H, we varied N_1 from 1 to 100 to find N_2 that satisfies the equation. TheFigure 3 displays N_2 values corresponding to each N_1. Additionally, we calculated power losses for each (N_1, N_2)

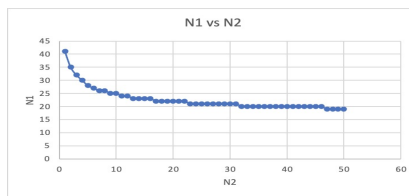

Fig. 3: N_1 vs N_2

pair and plotted the results. At $N_1 = 9$ and $N_2 = 25$, the

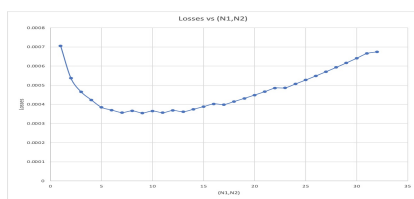

Fig. 4: Losses vs (N_1,N_2)

minimum power loss is 0.000354967W.

IV. Conclusion

The calculated mutual inductance (M from the provided equation) and associated values are accurate when coils are widely spaced. However, for overlapping coils (coils on top of each other), we need to derive more precise equations.

References

[1] Wireless Power Transfer for Electric Vehicles and Mobile Devices: Chun T. Rim; Chris Mi
[2] Analytical design of Archimedean spiral coils Kunwar Aditya